Keywords: LoRA • low-rank adaptation • Transformer attention • approximation theory

How Much Rank Does LoRA Need? Rank-Error Bounds for Transformer Attention

A Task-Dependent Theory Ties LoRA Rank to Downstream-Weighted Tail Energy — and Shows Softmax Saturation Can Shrink the Required Rank

By Gerard Conangla Planes

arXiv:2608.26052 • Preprint, August 2026

LoRA is the dominant paradigm for adapting large language models to downstream tasks, yet practitioners select the rank r by heuristic. This paper provides the first task-dependent theory of LoRA approximation error for Transformer attention: fixing a pretrained head, a target attention function, and a downstream input distribution, it bounds the minimum expected KL divergence achievable by a rank-r query LoRA update as an explicit function of the downstream-weighted tail energy of the target update matrix. A striking corollary is that softmax saturation can make the required rank strictly smaller than the rank needed to match the attention logits.

AT A GLANCE

Problem: LoRA rank r is chosen by trial-and-error despite being task-dependent; no theory predicts the KL error achievable at rank r.

Why it matters: Principled rank selection reduces parameter overhead while guaranteeing sufficient expressiveness for the adaptation.

Method: Bound the minimum expected KL error under rank-r query LoRA updates as a function of the downstream-weighted spectral tail energy T_r of the target update matrix.

Result: Softmax saturation lets rank needed to match the attention function be strictly lower than rank needed to match logits — constant separation for an explicit family.

Evidence: Theoretical bounds, extending to multi-head fused LoRA and joint query/key adaptation; construction proving tightness.

The Big Picture

LoRA (Low-Rank Adaptation) fine-tunes a pretrained weight matrix W_0 by adding $\Delta W = BA$ with $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$, keeping W_0 frozen. The rank r controls both the parameter count and the class of reachable adaptations. In current practice, r is chosen from a small grid $\{1, 2, 4, 8, 16\}$ by cross-validation, with no principled basis for the choice.

For Transformer attention, the query matrix W_Q determines the inner product structure of the attention mechanism. A rank-r update to W_Q shifts the key-query inner products in at most r directions. If the downstream task requires shifts in many independent directions, a small r is insufficient; if the task requires shifts in a low-dimensional subspace, a large r wastes parameters.

This paper asks: given the pretrained W_Q, the target W_Q^*, and the downstream input distribution $\mathcal{D}$, what is the minimum KL divergence between adapted and target attention distributions achievable by any rank-r query LoRA update?

Why Existing Approaches Fall Short

Existing analyses of LoRA operate at the level of weight-matrix approximation: a rank-r update approximates $\Delta W^* = W_Q^* - W_Q$ in spectral or Frobenius norm. But this misses two effects:

First, attention KL divergence depends on *downstream-weighted* approximation error. Components of ΔW^* that are large in matrix norm but project onto low-probability input directions have little effect on the adapted attention distribution. A matrix-norm-based analysis overestimates the required rank for these components.

Second, softmax applies a highly nonlinear map to the logits before computing the attention weights. Large logit differences saturate the softmax and make the attention distribution insensitive to further increases. A low-rank logit update can therefore achieve the same attention KL as a higher-rank update that matches the full logit change, because the relevant logit differences are already in the saturation regime.

Core Mechanism

The paper defines the **tail energy** T_r as the sum of eigenvalues beyond rank r in the downstream-weighted spectral decomposition of the target update:

$$T_r = \sum_{j>r} \lambda_j \left(\Sigma^{1/2} \Delta W^{*\top} \Delta W^* \Sigma^{1/2}\right)$$

where $\Sigma = \mathbb{E}_{\mathcal{D}}[xx^\top]$ is the downstream input covariance. The tail energy measures the fraction of the target adaptation that lies outside the subspace reachable by rank-r queries under the downstream distribution.

This quantity bridges weight-matrix approximation and attention KL divergence: a small T_r implies there exists a rank-r update with small expected KL error; a large T_r implies no rank-r update can achieve small error.

Technical Formulation

Fix an input x, query $q = W_Q x$, target query $q^* = W_Q^* x$, and key vectors $\{k_j\}$. The logit residual is $d = q - q^*$.

The KL divergence between the adapted and target attention distributions is:

$$\mathrm{KL}(\mathrm{softmax}(q^*) \| \mathrm{softmax}(q^* - \Delta q)) = \psi(\|d\|_2)$$

where ψ is an increasing function bounded above by $\min\{\|d\|_2^2/4, \sqrt{2}\|d\|_2\}$.

For attention probabilities bounded away from zero (condition $\pi_j \geq \gamma > 0$ for all j), the lower bound is:

$$\mathrm{KL} \geq \psi(\|d\|_2) \geq c_\gamma \|d\|_2$$

for an explicit constant c_γ depending on γ.

Under realizability, geometry, and moment conditions, the best rank-r expected KL error $\epsilon^*(r)$ satisfies:

$$c_1 \psi(\sqrt{T_r}) \;\leq\; \epsilon^*(r) \;\leq\; \min\left\{\tfrac{T_r}{4},\, \sqrt{2T_r}\right\}$$

with explicit constants c_1. These bounds are tight: the paper constructs families achieving both.

Learning or Inference Procedure

The theory yields a practical rank selection procedure:

1. Collect a sample $\{x_i\}$ from the downstream distribution
2. Estimate $\Sigma = \frac{1}{n}\sum_i x_i x_i^\top$
3. Compute the downstream-weighted SVD of the target update ΔW^* (estimated from a small fine-tuning probe if needed)
4. Evaluate T_r for each candidate rank r
5. Select r as the smallest rank with T_r below the desired KL error tolerance

At fine-tuning time, only the rank-r query LoRA is trained; all other weights remain frozen.

What the Guarantee Says

The bounds establish that the minimum KL error is controlled by T_r: if $T_r \leq \tau^2$, then a rank-r update can match the target attention to within $\min\{\tau^2/4, \sqrt{2}\tau\}$ expected KL. For saturated attention (large logit differences), the softmax nonlinearity allows this bound to be substantially tighter than the logit approximation error T_r alone would suggest.

The paper proves that there exist target functions where the attention-matching rank is a constant factor smaller than the logit-matching rank—closing a gap between empirical observations that low rank often suffices and the prior absence of theory explaining why.

Experimental Findings

The paper validates the rank-error bounds on two settings:

Synthetic attention heads: For planted low-rank targets, the measured KL error at each rank closely tracks $\min\{T_r/4, \sqrt{2T_r}\}$, confirming the upper bound is tight within a factor of 2.

LLM fine-tuning (RoBERTa-large, GLUE): The rank selected by the tail energy criterion achieves comparable task accuracy to a grid-searched rank in 8/9 GLUE tasks, while using 30–60% fewer LoRA parameters on average. On STS-B (a semantic similarity task where attention must capture fine-grained input directions), the criterion correctly predicts that rank 4 is insufficient and recommends rank 16.

Ablations and Interpretation

Softmax saturation effect: For heads with large pre-softmax logit magnitudes ($\|q^*\|_\infty > 10$), the attention-matching rank is consistently 2–4× lower than the logit-matching rank, validating the theoretical separation.

Multi-head fused LoRA: When a single shared B is used across all heads, T_r must aggregate over head-specific downstream covariances; the bound remains tight under this extension.

Joint query/key updates: The analysis generalizes to adapting both W_Q and W_K simultaneously, with T_r defined over the joint key-query interaction tensor.

Distribution shift: If $\mathcal{D}$ is misspecified (e.g., using a generic corpus when the downstream distribution is specialized), T_r may underestimate the required rank. Domain-specific calibration data is therefore recommended for rank estimation.

Reference

Gerard Conangla Planes. **How Much Rank Does LoRA Need? Rank-Error Bounds for Transformer Attention.** arXiv:2608.26052, August 2026. https://arxiv.org/abs/2608.26052

Keywords: constrained decoding • context-free grammar • structured generation • pushdown automata

Stay Within Your Bounds: Distance-Guided Decoding for Guaranteed Context-Free Grammar Compliance

Lookahead Distance to Acceptance Closes the Infeasibility Gap in Grammar-Constrained LM Decoding

By Vincenzo Collura, Karim Tit, Eleonora Giunchiglia, Mike Papadakis, Maxime Cordy

arXiv:2608.28229 • EMNLP 2026 Findings (Long Paper)

Grammar-constrained decoding guarantees that a language model's output satisfies a target context-free grammar—enabling reliable generation of JSON, SQL, and logical formulas. But existing methods fail silently: they allow prefixes that no valid completion can extend within the token budget. This paper fixes that gap with a lookahead-guided framework that computes upper-bound distances to the nearest accepting PDA state, uses them to prune infeasible tokens online, and guides beam search toward acceptance. Every output is guaranteed syntactically valid; completion quality improves across all evaluated structure types.

AT A GLANCE

Problem: Grammar masks block individually illegal tokens but permit prefixes from which no budget-feasible completion satisfies the grammar — producing invalid outputs despite the mask.

Why it matters: Hard syntactic guarantees are essential for code generation, structured data extraction, and symbolic planning.

Method: Precompute bounded PDA summaries with upper-bound distances to acceptance; prune tokens that exceed the remaining budget; bias beam search toward acceptance.

Result: 100% syntactic validity on JSON, SQL, and LTL benchmarks; improved completion quality over prior constrained baselines.

Evidence: Experiments across three structure types and multiple grammar sizes; ablations on budget sensitivity and beam width.

The Big Picture

Structured generation tasks require outputs that are syntactically valid instances of a target language: JSON objects, SQL queries, Python expressions, or formulas in linear temporal logic (LTL). Language models produce token sequences autoregressively, and without constraints they generate syntactically invalid outputs at a significant rate.

Grammar-constrained decoding solves this by tracking a pushdown automaton (PDA) derived from the target context-free grammar (CFG). At each step, it masks out tokens that are not consistent with the current PDA state, ensuring only locally valid tokens are sampled. This prevents any single token from violating the grammar.

But local validity is not global validity. A sequence of locally valid tokens can lead the PDA into a configuration from which no complete valid string can be generated within the remaining budget. The decoder emits a prefix that is syntactically valid so far, but cannot be completed.

Why Existing Approaches Fall Short

Two failure modes arise in prior grammar-constrained decoding:

Tokenizer-grammar mismatch: LM subword tokenizers may split strings at positions that do not align with grammar terminal boundaries. A token that spans multiple grammar symbols can leave the PDA in a non-standard mid-rule state that requires specific continuations — which the remaining token vocabulary may not support.

Budget exhaustion: Even with perfect tokenizer alignment, the PDA may enter a configuration requiring many more tokens to reach acceptance than remain in the budget. The decoder continues generating but cannot produce a valid closing sequence.

Both failure modes share the same root: the token mask checks local legality (is this token consistent with the current PDA state?) but not global feasibility (is there any valid completion within budget?).

Core Mechanism

The paper introduces a **distance oracle**: for each PDA configuration $c = (q, \gamma)$ (state q, stack content γ), precompute an upper-bound estimate $\hat{d}(c)$ of the minimum number of additional tokens needed to reach any accepting configuration.

At each decoding step with B tokens remaining, a token a is permitted only if:

1. It is locally consistent with the current PDA state (existing mask)
2. The successor configuration $c' = \delta(c, a)$ satisfies $\hat{d}(c') \leq B$

Condition 2 is the new feasibility filter. It ensures that after token a is emitted, the remaining budget is sufficient to reach acceptance.

Among the feasible tokens, beam search is guided by a distance-adjusted score:

$$s'(a) = \log p_\theta(a \mid x_{<t}) - \lambda \cdot \hat{d}(\delta(c, a))$$

which biases the beam toward tokens that reduce the distance to acceptance, improving completion quality.

Technical Formulation

Let $\mathcal{A} = (Q, \Sigma, \Gamma, \delta, q_0, F)$ be the PDA for the target CFG. Define the **distance to acceptance**:

$$d(c) = \min\Big\{ n \in \mathbb{N} : \exists w \in \Sigma^n,\ c \xrightarrow{w} (q_f, \varepsilon),\ q_f \in F \Big\}$$

with $d(c) = \infty$ if no such w exists.

The paper computes upper bounds $\hat{d}(c) \geq d(c)$ by summarizing PDA reachability with bounded lookahead:

1. For each PDA state q, compute the shortest path in the non-deterministic reachability graph over (q, γ) pairs to any $q_f \in F$, with stack operations truncated at a bounded depth.
2. Propagate distances bottom-up through grammar rule bodies.

The computation is done offline before decoding. Online, only a table lookup per (configuration, token) pair is required.

For a token budget T and a model generating tokens $a_1, \dots, a_T$:

$$a_t^* = \arg \max_{a:\, \hat{d}(\delta(c_t, a)) \leq T - t} s'(a)$$

guarantees that every emitted prefix can be completed within the remaining budget.

Learning or Inference Procedure

The framework operates purely at inference time:

1. Given a target CFG, compile it to a PDA and compute the distance oracle $\hat{d}(\cdot)$ offline.
2. At each decoding step, compute the set of feasible tokens (locally valid and budget-feasible).
3. Apply beam search with distance-guided scores.
4. Return the highest-scoring complete parse at the end.

No fine-tuning or modification of the base language model is required. The framework integrates with any autoregressive LM.

What the Guarantee Says

The decoder is **sound**: every generated sequence is accepted by the target grammar. Specifically, if the initial budget T satisfies $T \geq \hat{d}(q_0, \perp)$ (the oracle distance from the initial configuration), then at least one feasible token exists at every step and the decoder terminates with a valid output.

The distance oracle guarantee is: for every configuration c encountered during decoding, $\hat{d}(c) \geq d(c)$, so the feasibility filter never incorrectly prunes a token that leads to a valid completion.

Experimental Findings

Evaluated on three structure types with three LLMs (GPT-2-XL, LLaMA-3-8B, Mistral-7B-v0.3):

- **JSON:** 100% syntactic validity for all models and grammar sizes; schema match rate +9.3 pp over greedy and +6.1 pp over token-mask-only baselines.
- **SQL:** 100% parse validity; execution accuracy +7.8 pp over token-mask-only decoding on Spider benchmark.
- **LTL:** 100% formula well-formedness; satisfaction rate of generated plans +11.2 pp over prior constrained baselines.
- **Latency:** Oracle precomputation takes 0.3–1.8 s for practical grammar sizes; online overhead per step is negligible.

Ablations and Interpretation

Budget sensitivity: With tight budgets (T only 10% above $d(q_0, \perp)$), the feasibility filter prunes aggressively and occasionally leaves no feasible token at mid-sequence steps. A budget cushion of 30% eliminates this; the paper recommends budget estimation from grammar depth.

Distance-guidance weight λ: $\lambda = 0$ (pure language model score) recovers the token-mask baseline; $\lambda = 0.5$ achieves the best completion quality. Larger λ overconstrains generation toward acceptance and reduces fluency.

Beam width: Distance-guided beam search with width 4 outperforms greedy distance-guided decoding by +2.1 pp on JSON schema match, with diminishing returns beyond width 8.

Grammar size: The oracle scales well to grammars with up to 500 production rules, covering real SQL dialects. For larger grammars, the bounded-lookahead approximation introduces small oracle errors but validity is preserved.

Reference

Vincenzo Collura, Karim Tit, Eleonora Giunchiglia, Mike Papadakis, Maxime Cordy. **Stay Within Your Bounds: Distance-Guided Decoding for Guaranteed Context-Free Grammar Compliance.**

arXiv:2608.28229, EMNLP 2026 Findings. https://arxiv.org/abs/2608.28229

Keywords: vision-language models • assistive technology • salience • low vision

Focus Where It Counts: A Salience-Driven Vision-Language Model for Low Vision Assistance

Captioning That Prioritizes Scene Elements by Human-Centered Importance, Not Photographic Coverage

By Jiazhao Liang, Hao Huang, Shuaihang Yuan, et al.

arXiv:2608.28218 • ECCV 2026

Vision-language models offer powerful scene captioning but are designed for comprehensive coverage, not for the ordered priorities of users with blindness or low vision. This paper introduces Salience-LLaVA, a salience-driven VLM trained to mention scene elements in order of their functional importance for human-centered navigation and interaction. Supported by three newly curated salience-annotated datasets, the model places the highest-priority element first in 84.3% of test images and is rated 4.2/5 by blind and low-vision users—versus 3.1/5 for standard LLaVA.

AT A GLANCE

Problem: General-purpose VLMs describe images comprehensively but do not order information by functional relevance for users with blindness or low vision.

Why it matters: BLV users often hear only the first few seconds of a caption; placing important information last renders AI assistance unreliable for navigation and safety.

Method: Curate salience-annotated datasets; finetune LLaVA with a combined captioning and ranking loss to produce salienceordered descriptions; prepend salience scores as soft prompts.

Result: 84.3% top-1 salience ordering accuracy (vs. 51.2% for LLaVA); 4.2/5 user helpfulness rating vs. 3.1/5.

Evidence: Three datasets, user study with BLV participants, transfer experiments on VizWiz.

The Big Picture

Assistive technology for persons with blindness or low vision (BLV) is undergoing a rapid transformation as vision-language models replace earlier rule-based systems. VLMs can describe complex scenes in natural language, answer visual questions, and identify objects—capabilities that are transformative for BLV users navigating the world.

But VLMs are trained on captioning benchmarks that reward comprehensive coverage: a high CIDEr score requires mentioning many objects accurately. For a BLV user in a grocery store, a caption that begins with the color and pattern of the floor tiles before mentioning that the aisle is blocked by a forklift is not just unhelpful—it is dangerous.

This paper identifies the ordering problem as a first-class challenge in assistive VLM design and proposes a framework that addresses it through salience-aware training.

Why Existing Approaches Fall Short

General-purpose captioning: Models trained on MSCOCO and similar benchmarks describe images from top-left to bottom-right, or by object size/saliency in the photographic sense. Neither corresponds to functional importance for BLV navigation or interaction.

VQA-based assistance: Answering specific questions ("Is the crosswalk light green?") requires the user to know what to ask. Open-ended scene description is essential for unknown environments.

Prior assistive captioning: Work on VizWiz-focused models improves answer accuracy for BLV-generated images but does not address the ordering of information in free-form descriptions.

The key missing component is a notion of *salience* grounded in human-centered priorities rather than photographic composition.

Core Mechanism

The Salience-LLaVA framework has two components:

Salience scoring module: A lightweight MLP head added to the VLM's visual encoder that predicts a scalar salience score $s_i \in [0, 1]$ for each detected bounding box o_i. The module is trained with a point-wise ranking loss against human-annotated salience ranks.

Salience-conditioned caption generation: The caption generator receives the ranked object list $[o_{\pi(1)}, o_{\pi(2)}, \ldots]$ (sorted by predicted salience) as soft prompt tokens prepended to the image features. The LM is then trained to generate captions that mention objects in the order they appear in the ranked list.

At inference, salience scoring and caption generation run in sequence: score all detected objects, sort by salience, generate a caption conditioned on the sorted list.

Technical Formulation

Let $O = \{o_1, \ldots, o_n\}$ be the detected objects with ground-truth salience scores s_i^*. The salience head predicts $\hat{s}_i$ and the ranking loss penalizes misordered pairs:

$$\mathcal{L}_{\text{rank}} = \sum_{i,j:\, s_i^* > s_j^*} \max(0,\, \hat{s}_j - \hat{s}_i + \delta)$$

with margin $\delta = 0.1$.

The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda \mathcal{L}_{\text{rank}}$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy on the caption sequence y and $\lambda = 0.3$.

The implicit ordering $\hat{\pi}$ extracted from the generated caption is defined by the first mention position of each object:

$$\hat{\pi}(i) = \min\{t : o_i \text{ is mentioned at token position } t\}$$

The ranking loss targets the ordering induced by $\hat{s}$ matching s^*.

Learning or Inference Procedure

Dataset construction:

1. **Salience COCO:** 30k MSCOCO images; annotators rank all detected objects by importance for a BLV person navigating the scene.
2. **Salience Flickr:** 10k Flickr images with the same protocol.
3. **Salience VizWiz:** 15k images from BLV users' own photographs; salience ranks derived from crowd-sourced question answers.

Training: Fine-tune LLaVA-1.6 on Salience COCO and Salience Flickr for 3 epochs with $\lambda = 0.3$; evaluate zero-shot transfer to Salience VizWiz.

Inference: Run object detection (DINO-v2), score salience, sort, generate caption with sorted soft prompt. Total overhead: +38 ms per image on an A100.

What the Guarantee Says

The framework guarantees salience ordering in expectation over the training distribution: at convergence, the expected rank of the highest-salience object's first mention position approaches 1. The guarantee is probabilistic—individual captions may reorder objects due to LM fluency constraints—but the aggregate improvement is statistically significant ($p < 0.001$).

Experimental Findings

- **Top-1 salience ordering:** Salience-LLaVA 84.3% vs. LLaVA-1.6 51.2%, GPT-4V-captioning 58.7%.
- **VizWiz transfer:** Salience ordering F1 +12.3 pp over LLaVA-1.6 without VizWiz fine-tuning.
- **Caption quality:** SPICE +2.1, CIDEr +5.3 over LLaVA-1.6, confirming salience ordering does not hurt coverage.
- **User study (n=24 BLV participants):** Helpfulness 4.2/5 vs. 3.1/5 for LLaVA; 20/24 preferred Salience-LLaVA for navigation tasks; most cited "important info came first."

Ablations and Interpretation

Soft prompt vs. hard ordering: Providing the ranked object list as a hard prefix (forcing the caption to start with each object in order) achieves 100% top-1 ordering but reduces fluency (SPICE −8.4). Soft prompting achieves 84.3% ordering with only +2.1 SPICE gain.

Salience source: Using ground-truth salience ranks at inference yields 92.1% top-1 ordering, establishing an upper bound; the predicted salience scores are 93% rank-correlated with human rankings.

λ ablation: $\lambda = 0$ (no ranking loss) reduces top-1 ordering to 54.1%; $\lambda = 1.0$ improves ordering to 87.2% but reduces CIDEr by 11 points; $\lambda = 0.3$ balances both.

Domain specificity: Salience annotations vary across domains: in traffic scenes, moving objects (cars, cyclists) are highest salience; in indoor scenes, social agents (people) dominate. The model generalizes well within COCO categories but may require domain-specific annotation for specialized environments.

Reference

Jiazhao Liang, Hao Huang, Shuaihang Yuan, et al. **Focus Where It Counts: A Salience-Driven Vision-Language Model for Low Vision Assistance.** arXiv:2608.28218, ECCV 2026. https://arxiv.org/abs/2608.28218

Keywords: diffusion models • reinforcement learning • reward alignment • path-space gradient

Designing Reinforcement Learning for Diffusion Models: A Unified Path-Space View

Importance Sampling Between Sampling SDEs Connects All Major Diffusion-RL Algorithms Under One Principle

By Yixian Xu, Yuanrui Zhang, Shengjie Luo, Liwei Wang, Di He

arXiv:2608.14430 • Peking University; ByteDance Seed, August 2026

Reinforcement learning fine-tuning aligns diffusion models with task-specific rewards, but its algorithmic landscape is fragmented: reverse-trajectory methods, forward-matching methods, and value-gradient approaches each appear to follow different principles. This paper shows they all arise from a single path-space importance-sampling derivation applied to the diffusion SDE. The unification reveals a structured design space whose axes are value-gradient estimation, weight functions, and sampling strategy. From this space the authors derive a multi-sample KDE value-gradient estimator that reduces gradient variance 3.2× and outperforms prior methods by up to 8.3 pp on human preference tasks.

AT A GLANCE

Problem: RL methods for diffusion model alignment are algorithmically fragmented with no shared theoretical basis, preventing principled comparison or design.

Why it matters: A unified framework allows systematic ablation and derivation of new, more efficient algorithms.

Method: Importance sampling between sampling SDEs on trajectory space; derive policy gradient, then its value-gradient form; organize algorithm space by estimation, weight, and sampling choices.

Result: KDE value-gradient estimator cuts gradient variance 3.2× and improves human preference alignment +8.3 pp on SD3.5-M.

Evidence: Ablations over all design-space axes; evaluation on image generation (SD3.5-M) and vision-language alignment (Qwen-Image).

The Big Picture

Diffusion models define a generative process through a forward SDE that gradually adds noise to data and a reverse SDE that denoises from pure noise to samples. Fine-tuning for reward alignment means biasing the reverse process toward high-reward x_0 without destroying distributional coverage. Reinforcement learning provides the right framework: treat each denoising step as a policy action and optimize expected reward.

But several RL algorithms for diffusion models have been proposed independently, and their relationships are opaque:

- **DDPO/DPO-Diffusion:** Apply PPO-style clipped probability ratios over the full reverse trajectory.
- **AlignProp/ReFL:** Backpropagate reward gradients directly through the reverse SDE.
- **Forward-matching methods:** Train on reward-weighted forward-noised samples.

This paper asks: are these the same algorithm with different approximations, or genuinely different approaches?

Why Existing Approaches Fall Short

Without a unifying theory, practitioners must choose among algorithms without understanding what design choices actually matter. This has several consequences:

Ad hoc hyperparameter selection: Clipping thresholds, sampling schedules, and weight functions are set empirically; small changes produce large performance differences.

High variance: REINFORCE-style estimators applied to the full reverse trajectory have variance that scales with trajectory length. Prior methods apply heuristics to reduce variance without theoretical grounding.

Inaccessible algorithm space: It is unclear whether a new estimator can be derived within this framework or whether improvements require fundamentally new ideas.

Core Mechanism

The paper applies importance sampling between the current reverse-SDE measure $p_\theta(\tau)$ and the reward-weighted measure $p^*(\tau) \propto p_\theta(\tau)R(x_0)$ on the full trajectory space $\tau = (x_T, \ldots, x_0)$.

The importance weight is:

$$w(\tau) = \frac{p^*(\tau)}{p_\theta(\tau)} = \frac{R(x_0)}{\mathcal{Z}}$$

where $\mathcal{Z} = \mathbb{E}[R(x_0)]$ normalizes the measure.

The policy gradient $\nabla_\theta J(\theta) = \nabla_\theta \mathbb{E}_{p_\theta}[R(x_0)]$ becomes:

$$\nabla_\theta J = \mathbb{E}_{p_\theta}[R(x_0)\nabla_\theta \log p_\theta(\tau)]$$

This is a standard policy gradient on trajectory space. Factoring $\log p_\theta(\tau) = \sum_t \log p_\theta(x_{t-1}|x_t)$ shows that each reverse step contributes a per-step gradient weighted by the downstream reward.

Technical Formulation

The **value function** at denoising step t is:

$$V^\pi(x_t) = \mathbb{E}_{p_\theta(\tau_{t:0}|x_t)}[R(x_0)]$$

The policy gradient can be re-expressed in value-gradient form:

$$\nabla_\theta J = \mathbb{E}_{x_T}[\nabla_\theta V^\pi(x_T)]$$

where $x_T \sim \mathcal{N}(0, I)$ is the initial noise.

The paper proposes a **multi-sample KDE value estimator**:

$$\hat{V}(x_t) = \frac{\sum_{i=1}^{K} R(x_0^{(i)}) K_h(x_0 - x_0^{(i)})}{\sum_{i=1}^{K} K_h(x_0 - x_0^{(i)})}$$

where K rollouts $\{x_0^{(i)}\}$ are drawn from $p_\theta(\cdot|x_t)$ and a Gaussian kernel K_h smooths the estimate. Variance is:

$$\mathrm{Var}(\hat{V}_{\mathrm{KDE}}) = O(1/K)$$

compared to $O(1)$ for the single-sample REINFORCE baseline.

The three design axes of the unified space are:

1. **Weight function** $f(w(\tau))$: identity (vanilla PG), clip (ϵ-PPO), binary advantage ($\mathbb{1}[R > c]$)
2. **Value-gradient estimation**: direct backprop through SDE, REINFORCE, multi-sample KDE, separate critic
3. **Sampling strategy**: on-policy, off-policy with importance correction, mixed

Existing algorithms correspond to specific combinations; new ones emerge from unexplored combinations.

Learning or Inference Procedure

Training loop:

1. Sample M initial noises $\{x_T^{(m)}\}$
2. For each $x_T^{(m)}$, draw K rollouts of the full reverse SDE to obtain $\{x_0^{(m,k)}\}$
3. Compute reward $R(x_0^{(m,k)})$ for each sample
4. Compute KDE value estimates $\hat{V}(x_t^{(m)})$ at each step
5. Compute the value gradient and apply an optimizer step to θ

Inference: Standard diffusion model sampling; no change to the inference pipeline.

What the Guarantee Says

Under smoothness of R and the reverse SDE transition kernels, the KDE estimator satisfies:

$$\mathrm{Var}(\hat{V}_{\mathrm{KDE}}) \leq \frac{C\|R\|_\infty^2}{Kh^d}$$

where d is the data dimension and h the kernel bandwidth. Bias is $O(h^2)$. The optimal bandwidth balancing bias and variance gives $h = O(K^{-1/(d+4)})$, and the total mean-squared error decays as $O(K^{-4/(d+4)})$.

Experimental Findings

- **SD3.5-M (image generation):** KDE estimator outperforms single-sample REINFORCE by +8.3 pp on human preference alignment; outperforms AlignProp by +4.6 pp.
- **Qwen-Image (VQA alignment):** +4.1 pp absolute over AlignProp on target reward; +2.3 pp over DDPO.
- **Variance reduction:** KDE with $K = 8$ reduces gradient variance 3.2× vs. single-sample; variance continues decreasing to $K = 32$ with diminishing returns.
- **Design space ablations:** Binary advantage weighting degrades performance −3.1 pp vs. identity; off-policy with importance correction recovers +1.8 pp of the gap.

Ablations and Interpretation

Number of rollouts K: The best pass-at-1 reward improves monotonically with K up to $K = 16$; wall-clock time scales linearly. At $K = 8$ the compute-performance trade-off is near-optimal.

Weight function: Identity weighting (vanilla PG) outperforms PPO clipping in the low-data regime; clipping helps stability when off-policy data is reused.

Trajectory length: Methods that apply the gradient estimator only at the final denoising step (equivalent to treating the full reverse trajectory as a black box) underperform full-trajectory methods by −5.2 pp, confirming that per-step value estimates matter.

Softmax saturation (for discrete diffusion variants): The KDE estimator generalizes to discrete diffusion language models with a categorical kernel; variance reduction persists (2.8×).

Reference

Yixian Xu, Yuanrui Zhang, Shengjie Luo, Liwei Wang, Di He. **Designing Reinforcement Learning for Diffusion Models: A Unified Path-Space View.**

arXiv:2608.14430, Peking University; ByteDance Seed, August 2026. https://arxiv.org/abs/2608.14430

Keywords: adversarial robustness • mixture-of-experts • mode connectivity • multi-norm defense

Robust CurveMoE: Multi-Norm Adversarial Defense for Mixture-of-Experts Models via Mode Connectivity

Low-Loss Bézier Paths Through MoE Weight Space Combine Per-Norm Robust Endpoints Into a Single Multi-Norm Robust Model

By Xu Zhang, Ren Wang

arXiv:2608.26043 • Preprint, August 2026

Mixture-of-Experts models are increasingly deployed at scale, but their robustness to adversarial perturbations across multiple ℓ_p threat models simultaneously has been understudied. This paper proposes Robust CurveMoE: train separate MoE models robust under each threat norm, find low-loss Bézier curve paths connecting them through weight space, and sample a final model from the connecting path. The resulting model achieves multi-norm adversarial robustness without ensemble inference overhead, with clean accuracy dropping only 1.2% relative to the endpoint models.

AT A GLANCE

Problem: MoE models trained for robustness under one ℓ_p norm are vulnerable to other norms; union adversarial training is expensive and hurts accuracy.

Why it matters: Production systems face diverse adversarial attacks; a single model must defend across all without multiplying inference cost.

Method: Find Bézier curve paths through MoE weight space connecting per-norm robust endpoints; sample from the path at the point of best multi-norm trade-off.

Result: Robust accuracy [86.2%, 71.4%, 68.3%] under [ℓ_∞, ℓ_2, ℓ_1] vs. 4.8% clean accuracy drop for union AT; CurveMoE drops only 1.2%.

Evidence: ResNet MoE on CIFAR-100; ablations on path degree and Gumbel-softmax relaxation.

The Big Picture

Adversarial robustness is typically studied under a single threat model: ℓ_∞-bounded perturbations with radius ϵ_∞. But real-world adversaries can construct attacks in any ℓ_p geometry, and a model robust to ℓ_∞ attacks may be easily fooled by ℓ_2 or ℓ_1 perturbations.

Multi-norm robustness — simultaneous robustness to ℓ_∞, ℓ_2, and ℓ_1 threats — has been addressed for dense networks by union adversarial training or ensembles, but both approaches fail for Mixture-of-Experts models:

Union AT trains on the worst-case perturbation over all three norms simultaneously. This is computationally expensive and leads to an accuracy-robustness trade-off that worsens as the number of norms increases.

Ensembles combine predictions from three separately robust models, tripling inference cost. For MoE models with thousands of experts, ensemble inference is prohibitive.

Mode connectivity offers a third path: connect separately trained robust models through low-loss paths in weight space and sample a model from the path. For dense networks this has been demonstrated; for MoE models, the routing-dependent loss landscape poses new challenges.

Why Existing Approaches Fall Short

Single-norm AT: Provides strong robustness under the target norm but typically transfers poorly to orthogonal norms. A model ℓ_∞-trained at $\epsilon = 8/255$ may have robust accuracy below 40% under ℓ_2 attacks of matched strength.

Union AT for MoE: The multi-norm adversarial objective requires solving $\max_{p\in\{\infty,2,1\}} \max_{\|\delta\|_p \le \epsilon_p} \mathcal{L}(\theta; x+\delta, y)$ at each training step. For MoE models, the inner max interacts with discrete expert routing: the worst-case perturbation may change the routing path, invalidating the gradient from the current routing.

Mode connectivity for dense networks: Garipov et al. (2018) showed that dense network pairs trained independently can be connected by low-loss quadratic Bézier curves. For MoE models, the argmax routing creates discontinuities that break the Garipov construction; straight-line interpolation passes through high-loss regions.

Core Mechanism

The CurveMoE approach has four stages:

1. **Per-norm endpoint training:** Train θ_{ℓ_∞}, θ_{ℓ_2}, θ_{ℓ_1} using standard adversarial training under each norm respectively.
2. **Gumbel-softmax relaxation:** Replace argmax expert routing with Gumbel-softmax to make the loss differentiable with respect to weight interpolations along the connecting path.
3. **Path optimization:** Find a Bézier curve $\phi(t)$ connecting two endpoint models by minimizing the expected loss along the path under the relaxed routing.
4. **Path sampling:** Evaluate a grid of t values and select t^* that maximizes the minimum robust accuracy across all three norms on a held-out validation set.

For three endpoints, a triangular convex combination is used instead of a pairwise curve, providing simultaneous connectivity to all three robust models.

Technical Formulation

A degree-2 Bézier curve between θ_A and θ_B with midpoint θ_c is:

$$\phi(t) = (1-t)^2\theta_A + 2t(1-t)\theta_c + t^2\theta_B, \quad t \in [0,1]$$

The midpoint θ_c is optimized by minimizing:

$$\mathcal{L}_{\text{path}}(\theta_c) = \mathbb{E}_{t\sim U[0,1]} \sum_{p\in\{\infty,2,1\}} \alpha_p \mathcal{L}_p^{\text{adv}}(\phi(t))$$

where $\mathcal{L}_p^{\text{adv}}(\theta) = \max_{\|\delta\|_p \le \epsilon_p} \mathcal{L}(\theta; x+\delta, y)$.

For the three-endpoint triangular path with $t_1 + t_2 \le 1$:

$$\phi(t_1,t_2) = t_1\theta_{\ell_\infty} + t_2\theta_{\ell_2} + (1-t_1-t_2)\theta_{\ell_1}$$

with three midpoint parameters $\theta_{c_{12}}, \theta_{c_{13}}, \theta_{c_{23}}$ along each edge of the triangle.

The Gumbel-softmax relaxation at temperature τ_g:

$$\hat{g}_i = \frac{\exp((h_i+g_i)/\tau_g)}{\sum_j \exp((h_j+g_j)/\tau_g)}$$

where $g_i \sim \text{Gumbel}(0,1)$ and h_i are expert logits. During path optimization $\tau_g = 1.0$; reverted to hard routing ($\tau_g \to 0$) for evaluation.

Learning or Inference Procedure

1. Train θ_{ℓ_p} for each $p \in \{\infty, 2, 1\}$ using $\text{PGD} - 10$ adversarial training under the respective norm.
2. Initialize pairwise Bézier midpoints $\theta_c = (\theta_A + \theta_B)/2$.
3. Optimize θ_c for 50 epochs with Gumbel-softmax routing at $\tau_g = 1.0$, Adam optimizer, learning rate 10^{-3}.
4. Grid-search $t \in \{0, 0.05, \dots, 1.0\}$ on the validation set; select t^* maximizing $\min_p R_p(\phi(t^*))$ where R_p is robust accuracy under norm p.
5. The final model is $\phi(t^*)$ with hard routing; no inference overhead.

What the Guarantee Says

The path-loss objective guarantees that $\hat{V}(x_t)$ (the loss along the optimized path) is uniformly low across $t \in [0,1]$. Because the path connects models robust under each individual norm, and the path loss is low (implying robustness is maintained along the path), models at intermediate t values inherit multi-norm robustness properties from both endpoints.

This is not a formal robustness certificate but a constructive argument: the path endpoint models have certified robustness under their respective norms, and the path loss bounds how much that robustness degrades along the path.

Experimental Findings

Evaluated on ResNet-56 MoE with 8 experts on CIFAR-100, under PGD-50 attacks with $\ell_\infty \epsilon = 8/255$, $\ell_2 \epsilon = 0.5$, $\ell_1 \epsilon = 12$:

- **CurveMoE robust accuracy:** [ℓ_∞: 86.2%, ℓ_2: 71.4%, ℓ_1: 68.3%]
- **Best single-norm AT (per norm):** [ℓ_∞: 89.1%, ℓ_2: 74.2%, ℓ_1: 71.8%] (different models for each)
- **Union AT:** [ℓ_∞: 81.4%, ℓ_2: 68.7%, ℓ_1: 65.2%] with -4.8% clean accuracy drop
- **CurveMoE clean accuracy:** -1.2% relative to endpoint models
- **Path loss:** Maximum loss along the optimized path is 4% higher than at endpoints; linear interpolation paths peak at 31% higher, confirming the need for degree-2 curves.

Ablations and Interpretation

Path degree: Linear interpolation (degree 1) passes through high-loss regions for MoE models. Degree-2 Bézier curves find a low-loss path for all endpoint pairs tested. Degree 3 provides marginal improvement.

Gumbel temperature: $\tau_g = 1.0$ during path optimization gives smooth gradients; lower temperatures ($\tau_g = 0.1$) approximate hard routing but produce noisy gradients; $\tau_g = 0.5$ is a useful intermediate for larger MoEs.

Number of experts: Path optimization scales well to 16 and 32 experts; beyond 64 experts, path optimization requires more epochs to overcome routing instability.

Transfer to ViT-MoE: The approach transfers to Vision Transformer MoE architectures with token routing; mode connectivity persists, though the midpoint loss is 6% higher than for ResNet MoE, suggesting transformer MoE landscapes are less connected.

Reference

Xu Zhang, Ren Wang. **Robust CurveMoE: Multi-Norm Adversarial Defense for Mixture-of-Experts Models via Mode Connectivity.** arXiv:2608.26043, August 2026. https://arxiv.org/abs/2608.26043